Geethanjali College of Engineering and Technology (Autonomous)

Accredited by NAAC with A⁺ Grade; B.Tech. CSE, EEE, ECE accredited by NBA

Sy. No: 33 & 34, Cheeryal (V), Keesara (M), Medchal District, Telangana – 501301

Department of Computer Science and Engineering

Project – Progress Review Report (3rd Review) AY:25-26 April 2026

Date: _____

Batch Number:	4	**Section:**	CSE - G
Project Title:	MEDSCOPE		
Project Domain:	Software Engineering (Web Application Development)		
Project Guide & Designation:	G. Santhoshi, Assistant Professor		

Student Details

Sl. No.	Roll Number	Student Name	Contact Number	Signature & Date
1	22R11A05W7	M. Mounika	7396745069	
2	22R11A06W4	K. Priyanka	6300606482	
3	22R11A05V9	G. Bhavya Sree	8522061169	

1. Problem Statement (Brief)

Many users face difficulty in finding nearby pharmacies with available medicines, checking expiry dates, and receiving alerts for expiring medicines. There is no centralized system to manage pharmacy inventory, expiry alerts, and medicine availability across multiple pharmacies.

The proposed system MedScope provides a centralized platform where users can search medicines, pharmacies manage inventory, and admins monitor expiry alerts and analytics.

2. Objectives of the Project

- Develop a centralized medicine tracking system
- Provide real-time medicine availability
- Implement expiry alert notifications
- Enable pharmacy inventory management
- Provide role-based login (Admin, Pharmacy, User)
- Improve medicine accessibility
- Provide analytics dashboard

3. Project Status

Percentage of Project Completed (till Review 3): 85 %

Percentage of Project Work Pending (if any): 15%

3b. Status of Paper Publication

(Provide Journal / Conference details)

Not yet submitted / In progress

(Tick ✓ all appropriate items below)

4. System Architecture & Design Validation

Architecture Type: ☐ Monolithic ☐ Microservices ☐ Layered

Architecture Checks

- ☐ Clear separation (UI / Business Logic / DB)
- ☐ API design follows REST/standards
- ☐ Scalable design considered
- ☐ Loose coupling between modules

Architecture Diagram Finalized: ☐ Yes ☐ No

5. Technology Stack Proposed

Frontend	HTML,CSS,Bootstrap,Javascript	**Backend**	Python,Flask
Database	Firebase Firestore,CSV fallback	**Tools / APIs**	Firebase,Gmail,SMTP,APScheduler,Pandas,FPDF

6. Code Quality Review

Maintainability

- ☐ Clean, readable code
- ☐ Functions < 50 lines (good practice)
- ☐ No hardcoded values

Standards & Practices

- ☐ Follows coding standards (PEP8 / Java conventions etc.)
- ☐ Meaningful variable/function names
- ☐ Code comments where necessary

Advanced Checks

- ☐ DRY (No duplication)
- ☐ Proper error handling (try/catch)
- ☐ Logging implemented

7. Project Modules Identified

Module Name	Description	Assigned Member	% of Completion
User Module	Search medicines	M. Mounika	100%
Pharmacy Module	Inventory management	K. Priyanka	95%
Admin Module	Monitoring dashboard	G. Bhavya Sree	90%
Alert System	Expiry notifications	M. Mounika	90%

Integration Check

All modules integrated: ☐ Yes ☐ No **APIs/interfaces working:** ☐ Yes ☐ No

Data flow between modules verified: ☐ Yes ☐ No

Integration Issues (if any): _____

8. Working Prototype (Deep Verification)

System Type: ☐ Web ☐ Mobile ☐ Desktop ☐ ML ☐ IoT

Functional Flow Validation

Feature	Input Given	Output Observed	Status
Search Medicine	Enter medicine name	Displays available pharmacies and medicine details	☐ Pass ☐ Fail
User Login	Valid username & password	User successfully logged into dashboard	☐ Pass ☐ Fail
Expiry Alert System	Medicines with expiry date stored	Expiry notification/email alert generated	☐ Pass ☐ Fail
Dashboard Analytics	Admin login	Dashboard shows inventory and expiry statistics	☐ Pass ☐ Fail

End-to-End Workflow Executed: ☐ Yes ☐ No **Real Data Used:** ☐ Yes ☐ No

Deployment

☐ Local ☐ Cloud ☐ Hybrid

URL / Access Details: http://localhost:5000/

9. Code Quality Audit (Detailed)

Code Structure

☐ Proper modularization (functions/classes)

☐ Separation of concerns (UI/Logic/DB)

☐ Reusability implemented

Coding Standards

☐ Naming conventions followed

☐ Indentation & formatting proper

☐ No redundant/duplicate code

Complexity & Optimization

☐ Efficient algorithms used

☐ Time/space complexity considered

☐ Code optimized for performance

Version Control

☐ Git used

☐ Regular commits present

☐ Meaningful commit messages

Repository Link: https://github.com/Mounika-2303/MEDSCOPE

10. Testing & Validation (Deep Checks)

Testing Coverage

- ☐ Unit Testing
- ☐ Integration Testing
- ☐ System Testing
- ☐ User Acceptance Testing

Test Case Coverage

Total Test Cases: _15__ **Covered Modules (%):** __85____ %

Sample Test Case

Test ID	Scenario	Input	Expected	Actual	Status
TC_01	Search for medicine	Enter "Paracetamol"	List of pharmacies showing availability	Correct list displayed	☐ Pass ☐ Fail
TC_02	Valid login	Correct username & password	User redirected to dashboard	Login successful	☐ Pass ☐ Fail
TC_03	Add new medicine	Enter medicine details	Medicine added to database	Successfully added	☐ Pass ☐ Fail
TC_04	Database connectivity	Load application	Connect to Firebase	Connected successfully	☐ Pass ☐ Fail

11. Debugging & Issue Tracking

Issues Log

Issue	Root Cause	Fix Applied	Status
Firebase connection failure	Incorrect API key / configuration	Updated Firebase config and credentials	☐ Resolved ☐ Pending
Expiry alert not triggering	Scheduler not initialized correctly	Fixed APScheduler setup	☐ Resolved ☐ Pending
Medicine search delay	Inefficient query handling	Optimized search logic and indexing	☐ Resolved ☐ Pending
Email notification not sent	SMTP configuration issue	Corrected Gmail SMTP settings	☐ Resolved ☐ Pending

Debugging Approach

- ☐ IDE Debugger
- ☐ Logs/Tracing
- ☐ Breakpoints
- ☐ Monitoring Tools

Improvements After Debugging

- ☐ Performance improved
- ☐ Errors reduced
- ☐ Stability increased

12. Performance & Security Checks

Performance

Avg Response Time: __2 seconds__

Load Handling: ☐ Tested ☐ Not Tested **Bottlenecks Identified:** ☐ Yes ☐ No

Security

☐ Input validation
☐ Authentication implemented
☐ Authorization controls
☐ Data encryption (if applicable)

13. Documentation Verification

Document	Prepared	Verified	Remarks
SRS	☐	☐	
Design (UML/Architecture)	☐	☐	
Algorithm/Flowcharts	☐	☐	
Database Schema	☐	☐	
Final Report	☐	☐	
User Manual	☐	☐	

14. Presentation & Communication

☐ Problem clearly defined
☐ Methodology explained
☐ Architecture diagram included
☐ Results & outputs shown
☐ Conclusion & future scope included

15. Demo Readiness

☐ Live demo stable
☐ Backup demo available
☐ All members can explain

16. Individual Contribution (Strict Check)

Member	Module Handled	Lines/Work Done	Explanation Ability
M.Mounika	Backend Development, Expiry Alert System, Email Notification, API Integration	Backend logic, Flask routes, alert scheduler, email integration	☐ Good ☐ Avg ☐ Poor
K.Priyanka	Frontend Development, UI Design, Dashboard Pages, Forms	HTML, CSS, Bootstrap UI, search pages, dashboard UI	☐ Good ☐ Avg ☐ Poor
G.Bhavya Sree	Database Management, Firebase Integration, Data Handling, Testing	Firestore setup, data models, testing, bug fixes	☐ Good ☐ Avg ☐ Poor

17. Innovation & Impact

Novel Contribution: Real-time pharmacy medicine tracking with expiry alerts
Comparison with Existing Systems: Existing systems lack centralized expiry monitoring
Societal/Industry Relevance: Improves medicine availability and reduces wastage

18. Final Viva Readiness (Critical)

Check	Status
Project fully functional	☐ Yes ☐ No
All modules tested	☐ Yes ☐ No
Bugs resolved	☐ Yes ☐ No
Documentation completed	☐ Yes ☐ No
Demo stable	☐ Yes ☐ No

19. Mandatory Declarations

☐ Project is original work (no plagiarism)
☐ Code is understood by all team members
☐ System is ready for external evaluation
☐ Project documentation sample copy verified by guide
☐ Project exhibited at competitions

20. Additional Artifacts (Url to be mentioned)

GitHub: https://github.com/Mounika-2303/MEDSCOPE
Demo Video: https://drive.google.com/file/d/1C4lZNrAloVJzif6CEQtLQpCsI7oQiSZh/view?usp=sharing
Deployment Link: http://127.0.0.1:5000/

21. Reviewer (Internal Guide) Observations

✓ **Strengths:**

✓ **Weaknesses:**

✓ **Final Suggestion Before Viva:**

22. PO / PEO / CEP Mapping (to be marked)

Program Outcomes (PO): PO1 [ ] PO2 [ ] PO3 [ ] PO5 [ ] PO10 [ ]
Program Educational Objectives (PEO): PEO1 [ ] PEO2 [ ] PEO3 [ ]
Complex Engineering Problem (CEP): Research-based [ ] Design under constraints [ ] Modern tools [ ] Startup/Incubation scale [ ]

Declaration

We declare that the above information is true to the best of our knowledge.

Student Signatures:

1. _____ 2. _____ 3. _____ 4. _____

Internal Guide's Remarks and Recommendations for final Viva:

Project Guide Name: _____

Designation: _____

Signature (with Date): _____

Project Coordinator 1	Project Coordinator 2
Name: _____	Name: _____
Designation: _____	Designation: _____
Signature with date: _____	Signature with date: _____

Remarks of the Project Panel (if any):